

Fact #1 If $A \subseteq \mathbb{R}$ is Lebesgue measurable and $\lambda^*(A) > 0$ then $\exists E \subset A$ s.t. E is not measurable.

The proof of this uses inner/outer regularity of λ^N ... later!

So let $E \subseteq \mathbb{Q}(\mathbb{C})$ be a non-measurable set.

Fact #2 In general, the image of a Borel set under a continuous function is not a Borel set, and the image of a Lebesgue set under a continuous function may not be measurable.

E.g., there exists $E \subseteq \mathbb{R}$ Lebesgue measurable s.t. $E+E$ is not Lebesgue measurable.

However, if $f: I \rightarrow \mathbb{R}$ is continuous, $I \subseteq \mathbb{R}$ interval, $E \subset I$ Borel $\Rightarrow f(E)$ is Lebesgue measurable.

This can be generalized as follows:

Let X be a complete, separable metric space, $f: X \rightarrow \mathbb{R}$ continuous. If $B \subseteq X$ is a Borel set then $f(B) \subset \mathbb{R}$ is Lebesgue measurable (but not necessarily Borel).

end lect # 2

So by Fact #1 let $E \subseteq \mathbb{Q}(\mathbb{C})$ be non-measurable.

Let $F := g^{-1}(E) \subseteq \mathbb{C}$.

By the completeness of the Lebesgue measure, $\lambda^1(\mathbb{C}) = 0$

$\Rightarrow F$ is Lebesgue measurable and $\lambda^1(F) = 0$.

claim F is not a Borel set

let $h := g^{-1}: [0, 2] \rightarrow [0, 1]$ continuous.

Fact #3: The inverse image of a Borel set under a continuous function is a Borel set (proved below).

So, if F was Borel then $h^{-1}(F) = E$ would be Borel. $\times$
(recall, E is not even measurable!).

We now elaborate more on Facts #2 and #3.

Fact #2. Let $E \subset [0, 1]$ be a set that is not measurable (see 3.15 (a)). Consider $F := E \times \{1\} \subset \mathbb{R}^2$.

Then $F \subseteq [0, 1] \times \{1\}$ that has $\mathbb{L}^2$ measure zero, since

$$F \subseteq [0, 1] \times (1-\varepsilon, 1+\varepsilon) \quad \forall \varepsilon > 0 \quad \text{and (see Prop 2.9)}$$

$$(\mathbb{L}^2)^*(F) = \mathbb{L}([0, 1] \times (1-\varepsilon, 1+\varepsilon)) = 2\varepsilon.$$

Let $\varepsilon \rightarrow 0^+$ and conclude that $(\mathbb{L}^2)^*(F) = 0$.

Further, $[0, 1] \times \{1\}$ is closed, hence it is a Borel set.

By the completeness of $\mathbb{L}^2$, it follows that F is Lebesgue measurable with Lebesgue measure zero.

Consider the continuous projection

$$\pi: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$(x, y) \mapsto x.$$

Then $\pi(F) = E$ which is not measurable, so

F Lebesgue measurable + π continuous $\nRightarrow \pi(F)$ Lebesgue measurable.

Fact #3 let $I \subseteq \mathbb{R}$ be an interval and let $h: I \rightarrow \mathbb{R}$ be a continuous function.

claim if $B \subseteq \mathbb{R}$ is a Borel set, then $h^{-1}(B)$ is a Borel set
[see Remark 8.4 (v)]

Define $\mathcal{M} := \{E \subseteq \mathbb{R} : h^{-1}(E) \text{ is a Borel set}\}$

Since h is continuous, if $U \subseteq \mathbb{R}$ is open then $h^{-1}(U)$ is relatively open ($h^{-1}(U) = V \cap I$ for some open set V , in case I is not open),

and thus $U \in \mathcal{M}$.

(4.4)

Now $\mathcal{M}$ is a σ -algebra, because if $E \in \mathcal{M}$ then $h^{-1}(\mathbb{R} \setminus E) = I \setminus h^{-1}(E)$ still a Borel set, and if $E_n \in \mathcal{M}$ then $h^{-1}(\bigcup_{n=1}^{\infty} E_n) = \bigcup_{n=1}^{\infty} h^{-1}(E_n)$ still a Borel set.

So $\{ \text{open sets of } \mathbb{R} \} \subset \mathcal{M}$ ^{by (4.4),} therefore

$$\mathcal{B}(\mathbb{R}) \subset \mathcal{M} \text{ (}\sigma\text{-algebra),}$$

and this proves the claim.

Note that this continues to hold for a function $h: B \rightarrow Y$, continuous, $B \subseteq X$ Borel set, X and Y topological spaces.

Prop 4.13

By Example 3.15 a) and Example 4.11 we have

$$\mathcal{B}(\mathbb{R}^N) \subsetneq \{ \text{Lebesgue measurable sets} \} \subsetneq \mathcal{P}(\mathbb{R}^N)$$

we will see later on that

$$\mathcal{L}(\mathbb{R}^N) := \{ \text{Lebesgue measurable sets} \} = \mathcal{B}(\mathbb{R}^N) + \mathcal{N}$$

$$\text{where } \mathcal{N} := \{ A \subseteq \mathbb{R}^N : A \in \mathcal{L}(\mathbb{R}^N) \text{ and } \mathcal{L}^N(A) = 0 \}.$$

(see Thm 6.18)

5. UNIQUENESS

Thm 5.1 If μ is a measure on $(\mathbb{R}^N, \mathcal{L}(\mathbb{R}^N))$
(or $(\mathbb{R}^N, \beta(\mathbb{R}^N))$) and $\mu(R) = l(R) \quad \forall$ rectangles R
then $\mu(A) = l^N(A)$ for all $A \in \mathcal{L}(\mathbb{R}^N)$ (or $A \in \beta(\mathbb{R}^N)$).

[Compare with Proposition 2.14]

Proof step 1 we prove that $\mu \leq l^N$.

let $A \in \mathcal{L}(\mathbb{R}^N)$, and let $A \subseteq \bigcup_{n=1}^{\infty} R_n$, R_n rectangles.

$$\begin{aligned} \text{Then } \mu(A) &\leq \sum_{n=1}^{\infty} \mu(R_n) \Rightarrow \sum_{n=1}^{\infty} l^N(R_n) \\ &\quad \downarrow \text{monotonicity + subadditivity} \quad = \sum_{n=1}^{\infty} l(R_n) \quad (\text{Prop 2.11}) \end{aligned}$$

and taking the infimum over all such coverings, we get
 $\mu(A) \leq (l^N)^*(A) = l^N(A)$.

step 2 we prove that $l^N \leq \mu$ on bounded sets

let $A \in \mathcal{L}(\mathbb{R}^N)$ be bounded, say $A \subseteq R$ some rectangle R .

Then by step 1

$$\mu(R \setminus A) \leq l^N(R \setminus A) \Leftrightarrow \mu(R) - \mu(A) \leq \underbrace{l^N(R)}_{l^N(R)} - l^N(A)$$

and so $l^N(A) \leq \mu(A)$.

step 3 For arbitrary $A \in \mathcal{L}(\mathbb{R}^N)$, write $A = \bigcup_{n=1}^{\infty} A \cap (-n, n)^N$.

By Proposition 4.6 we have

$$\begin{aligned} \mu(A) &= \lim_{n \rightarrow \infty} \mu(A \cap (-n, n)^N) \quad (\text{step 2}) \\ &= \lim_{n \rightarrow \infty} l^N(A \cap (-n, n)^N) = l^N(A). \quad \blacksquare \end{aligned}$$

Now in an abstract setting...

let X be a nonempty set, let $\mathcal{M}$ be a σ -algebra on X , and let $\mathcal{E} \subseteq \mathcal{M}$. let μ, ν be measures on $(X, \mathcal{M})$ such that $\mu(X) = \nu(X) < +\infty$ and $\mu = \nu$ on $\mathcal{E}$.

Is it true that $\mu = \nu$ on $\sigma(\mathcal{E})$? (5.1)

The answer, in general, is No!

Consider $\mathcal{E} := \{A, B\}$ with $\mu(A) = \nu(A)$, $\mu(B) = \nu(B)$. we could have $\mu(A \cap B) \neq \nu(A \cap B)$.

Proposition 5.2 (5.1) holds if $\mathcal{E}$ is closed w.r.t. finite intersections.

"Proof" let $\mathcal{M} := \{A \in \mathcal{M} : \nu(A) = \mu(A)\}$.

clearly $\mathcal{E} \subseteq \mathcal{M}$, and if $\mathcal{M}$ is a σ -algebra then $\sigma(\mathcal{E}) \subseteq \mathcal{M}$, proving (5.1).

- (i) $\emptyset \in \mathcal{M}$ because $\nu(\emptyset) = 0 = \mu(\emptyset)$. $\mu(X) = \nu(X) < +\infty$
 (ii) If $A \in \mathcal{M}$ then $\mu(X \setminus A) = \mu(X) - \mu(A) = \nu(X) - \nu(A) = \nu(X \setminus A)$

so $X \setminus A \in \mathcal{M}$. (5.2)

- (iii) If $A_n \in \mathcal{M}$ and $A_n \subseteq A_{n+1}$, $n \in \mathbb{N}$, then by Proposition 4.6
- $$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n) = \lim_{n \rightarrow \infty} \nu(A_n)$$

$$= \nu\left(\bigcup_{n=1}^{\infty} A_n\right),$$

and so $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$.

To finish the proof, we would need to extend (iii)

to an arbitrary family $\{A_n\} \subseteq \mathcal{R}$.

Let us first remark that (ii) can be extended as follows:

(ii)', If $A, B \in \mathcal{R}$ and $A \subseteq B$ then $B \setminus A \in \mathcal{R}$.

$$\text{Indeed, } \mu(B \setminus A) = \mu(B) - \mu(A) \quad (*)$$

$$= \nu(B) - \nu(A) = \nu(B \setminus A),$$

where we use the fact that $\mu(A) = \nu(A) \leq \mu(X) = \nu(X) < +\infty$ in (*).

So the proof of Prop 5.2 actually follows from a more general context

Def 5.3 We say that $\Pi \subseteq \mathcal{P}(X)$ is a Π -system if

(i) $\Pi \neq \emptyset$,

(ii) $A, B \in \Pi \Rightarrow A \cap B \in \Pi$.

Def 5.4 We say that $\Lambda \subseteq \mathcal{P}(X)$ is a Λ -system if

(i) $X \in \Lambda$,

(ii) $A, B \in \Lambda, A \subseteq B \Rightarrow B \setminus A \in \Lambda$,

(iii) $\{A_n\} \subseteq \Lambda, A_n \subseteq A_{n+1} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \Lambda$.

N.B. σ -algebras are both Π -systems and Λ -systems, and conversely see Prop 5.7 (ii).

Thm 5.5 (Dynkin)

If Π is a Π -system, and Λ is a Λ -system, then

$$\Pi \subseteq \Lambda \Rightarrow \sigma(\Pi) \subseteq \Lambda.$$

We can now reformulate Prop 5.2 as

Corollary 5.6

If μ, ν are finite measures that agree on Π (Π -system) and $X \in \Pi$, then $\mu = \nu$ on $\sigma(\Pi)$.

Proof of Corollary 5.6

Let $\Lambda := \{A \in \mathcal{P}(X) : \nu(A) = \mu(A)\}$.

Since $X \in \Pi$, $\Lambda = \mathcal{P}(X)$ (as in the "proof") is a Λ -system in view of (i), (ii)' and (iii) in the proof of Prop 5.2.

Since $\Pi \subseteq \Lambda$, by Dynkin's Thm we have $\sigma(\Pi) \subseteq \Lambda$. ■

Remark 5.7

(i) The intersection of an arbitrary family of λ -systems

is a λ -system. Given $\mathcal{C} \subseteq \mathcal{P}(X)$, $\lambda(\mathcal{C})$ is the λ -system generated by $\mathcal{C}$.

(ii) If Λ is both a λ -system and a Π -system then it is a σ -algebra.

Indeed, by Def 5.4 (i) + (ii)

$$\Phi = X \setminus X \in \Lambda.$$

Also by (i) + (ii), if $A \in \Lambda \Rightarrow X \setminus A \in \Lambda$. (5.1)

Finally, if $\{A_n\} \subseteq \Lambda$, then

$$A_n' := \bigcup_{k=1}^n A_k \in \Lambda \text{ because } \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n', \quad A_n' \subseteq A_{n+1}',$$

$$X \setminus A_n' = \bigcap_{k=1}^n (X \setminus A_k) \in \Lambda \text{ because it is a } \Pi\text{-system} \quad (+ \text{ induction})$$

$\in \Lambda \text{ by (5.1)}$

Hence, again by (5.1) we have that $A_n' \in \Lambda$, and

by Def 5.4 (iii) we conclude that

$$\bigcup_{n=1}^{\infty} A_n' \in \Lambda.$$

end lect #8

Proof of Thm 5.5 [Dynkin Thm]

claim 1 $\sigma(\Pi) \subseteq \lambda(\Pi) \subseteq \Lambda$

N.B. σ -algebras are λ -systems

$\Rightarrow \lambda(\Pi) \subseteq \sigma(\Pi)$ always!

The last inclusion is trivial since $\Pi \subseteq \Lambda$. Recall that

$\lambda(\pi)$ is the smallest λ -system that contains π .

claim 2 $\lambda(\pi)$ is a σ -algebra.

If so, since $\pi \subseteq \lambda(\pi) \Rightarrow \sigma(\pi) \subseteq \lambda(\pi)$, proving claim 1.

Now by Remark 5.7 (ii), if $\lambda(\pi)$ is a π -system then it is a σ -algebra. So we are "reduced" to proving that $\lambda(\pi)$ is a π -system.

Def 5.3 (i) * By definition of λ -system, $X \in \lambda(\pi)$ hence $\lambda(\pi) \neq \emptyset$ (see Def 5.3).

let $C \in \lambda(\pi)$ and let $\Lambda_C := \{A \in \lambda(\pi) : A \cap C \in \lambda(\pi)\}$.

Note that $\lambda(\pi)$ is a π -system (if)

claim 3 $\Lambda_C = \lambda(\pi)$.

because given $C \in \lambda(\pi)$ and $A \in \lambda(\pi) \Rightarrow A \cap C \in \lambda(\pi)$

clearly, $\Lambda_C \subseteq \lambda(\pi)$. Conversely, (if) Λ_C is a λ -system $\uparrow$ claim 4
 $\pi \subseteq \Lambda_C \uparrow$ claim 5

then $\lambda(\pi) \subseteq \Lambda_C$, proving claim 3.

claim 4 Λ_C is a λ -system

(i) $X \in \lambda(\pi)$, and $X \cap C = C \in \lambda(\pi)$, so $X \in \Lambda_C$.

(ii) let $A, B \in \Lambda_C$, $A \subseteq B$.

Then $(B \setminus A) \cap C = (B \cap C) \setminus (A \cap C) \in \lambda(\pi)$ (it is a λ -system).
 $\underbrace{\quad}_{\in \lambda(\pi)} \quad \underbrace{\quad}_{\in \lambda(\pi)}$

(iii) let $A_n \in \Lambda_C$, $A_n \subseteq A_{n+1}$, $n \in \mathbb{N}$.

Then

$(\bigcup_{n=1}^{\infty} A_n) \cap C = \bigcup_{n=1}^{\infty} (A_n \cap C) \in \lambda(\pi)$ (it is a λ -system)
 $\underbrace{\quad}_{\in \lambda(\pi)} \quad \underbrace{\quad}_{\in \lambda(\pi) \text{ and increasing}}$